

DIVISION ALGORITHM

$$a = b \cdot q + r$$

Theorem: The Division Algorithm

$\forall a, b \in \mathbb{Z}$, with $b > 0$, $\exists!$ $q, r \in \mathbb{Z}$ such that

$$a = b \cdot q + r \quad \text{with } 0 \leq r < b$$

q quotient, r remainder.

Example: $a = 21, b = 2 \rightarrow 21 = 2 \cdot \underline{10} + \underline{1}$

proof: consider $S := \{a - bx \mid x \in \mathbb{Z}, a - bx \geq 0\}$

example: $a = 12, b = 5$

x	$12 - 5x$
0	12
1	7
2	2
3	-3
$\vdots$	$\vdots$

$\rightarrow S = \{2, 7, 12, \dots\}$

$\downarrow \& S$

claim: $S \neq \emptyset$

case 1: $a \geq 0$ ($x = 0$)

$$a - b \cdot 0 = a \in S$$

case 2: $a < 0$, set $x = a$

$$a - ba = \underset{\uparrow 0}{a} (\underset{\uparrow 0}{1} - b) \geq 0$$

$$\Rightarrow a - ba \in S$$

Let $r = \min(S)$, q is the corresponding x -value.

$$r = a - bq \Rightarrow a = bq + r$$

Towards a contradiction suppose that $r \geq b$

$$\begin{aligned} \Rightarrow r = a - bq \geq b &\Rightarrow r - b = a - b(q+1) \\ &\geq b - b = 0 \end{aligned}$$

$$\Rightarrow r - b \in S \text{ but } r - b < r = \min(S) \quad \text{W!}, *$$

Suppose that q, q' and r, r' are such that

$$a = bq + r = bq' + r'$$

$$\rightarrow bq + r = bq' + r' \quad . \text{ Assume } r' \geq r.$$

$$\rightarrow bq - bq' = r' - r$$

$$b(q - q') = r' - r$$

$$\begin{aligned} &\text{"} \quad \hookrightarrow 0 \leq r' - r < b \\ &k \cdot b \end{aligned}$$

$$\Rightarrow r' - r = 0 \Rightarrow r = r' \text{ and } q = q'.$$

□